Enhanced Object Tracking Based on an Improved CSRT Framework with Kalman Filtering and Deep Feature Refinement

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Abstract

Object tracking remains a challenging problem in computer vision due to factors such as appearance variation, occlusion, scale change, and background clutter. Among classical online trackers, the Channel and Spatial Reliability Tracker (CSRT) has demonstrated strong robustness by incorporating spatial reliability masks and discriminative correlation filters. However, the standard CSRT framework relies on handcrafted features and fixed update strategies, which limit its adaptability under fast motion and long-term tracking scenarios.

In this paper, we propose an enhanced CSRT-based tracking framework that improves robustness and stability without sacrificing real-time performance. The proposed method focuses on refining the core update mechanism of CSRT by integrating a motion-aware state estimation module and a lightweight deep regression component for scale and aspect ratio refinement. Specifically, the correlation response produced by CSRT is exploited to provide reliable target localization, while a motion prediction model is employed to mitigate drift and handle abrupt target displacement. In addition, a compact deep regression network is introduced to estimate target scale parameters within a constrained search region, avoiding the overhead of full-frame object detection.

Extensive experiments conducted on standard tracking benchmarks demonstrate that the proposed approach achieves superior performance compared to the original CSRT tracker, particularly in challenging scenarios involving fast motion, partial occlusion, and background distractors. The results show consistent improvements in tracking accuracy and stability while maintaining computational efficiency, making the proposed method suitable for real-time applications.

Keywords: Object Tracking, CSRT, Correlation Filter, Motion Estimation, Deep Regression, Kalman Filtering.

1 Introduction

Visual object tracking is a fundamental problem in computer vision, aiming to estimate the position and scale of a target object throughout a video sequence given its initial state. Despite significant progress in recent years, robust tracking in real-world scenarios remains challenging due to factors such as fast motion, occlusion, scale variation, and background clutter. In real-time tracking applications, these challenges are further amplified by strict computational constraints, which limit the use of heavy detection-based frameworks.

Among classical online tracking methods, correlation filter–based trackers have attracted considerable attention due to their favorable balance between accuracy and efficiency. In particular, the Channel and Spatial Reliability Tracker (CSRT) introduces spatial reliability masks and channel-wise weighting to enhance robustness against background interference. While CSRT demonstrates strong localization capability under moderate conditions, it still suffers from performance degradation in scenarios involving abrupt target motion, long-term drift, and inaccurate scale estimation, mainly due to its reliance on local response maximization and handcrafted feature representations.

To address these limitations, we propose an enhanced CSRT-based tracking framework that integrates motion-aware state estimation and deep feature refinement. A Kalman Filter is incorporated to model the temporal dynamics of the target, providing a principled motion prediction that stabilizes target localization under fast motion and partial occlusion. In addition, deep features are employed within a constrained search region to refine target scale and appearance representation, complementing the correlation response generated by CSRT without introducing the overhead of full-frame object detection.

The proposed approach preserves the efficiency and simplicity of the original CSRT tracker while significantly improving robustness and tracking stability. Extensive experiments on standard tracking benchmarks demonstrate that the proposed method outperforms the baseline CSRT tracker, particularly in challenging scenarios such as fast motion, scale variation, and background distraction. These results confirm the effectiveness of combining correlation filter–based tracking with Kalman filtering and deep feature refinement in a unified and efficient framework.

2 Related Works

2.1 Correlation Filter-Based Tracking

Correlation filter–based trackers have been widely studied due to their high efficiency and competitive accuracy in online visual tracking. Early methods such as

MOSSE and KCF formulate tracking as a ridge regression problem in the Fourier domain, enabling fast target localization through response map maximization. Subsequent works have extended this formulation by incorporating multi-channel features, spatial regularization, and scale adaptation to improve robustness under challenging conditions.

Among these approaches, the Channel and Spatial Reliability Tracker (CSRT) introduces spatial reliability masks and channel-wise weighting to suppress background interference and enhance discriminative power. By estimating both spatial and channel reliability, CSRT significantly improves tracking robustness compared to earlier correlation filter trackers. However, as a purely response-driven method, CSRT still relies on local maxima for target localization, which may lead to drift under fast motion, occlusion, or abrupt appearance changes. An overview of the CSRT framework is presented in Figure 1, highlighting the learning stage with spatial constraints and the localization stage based on weighted filter responses.

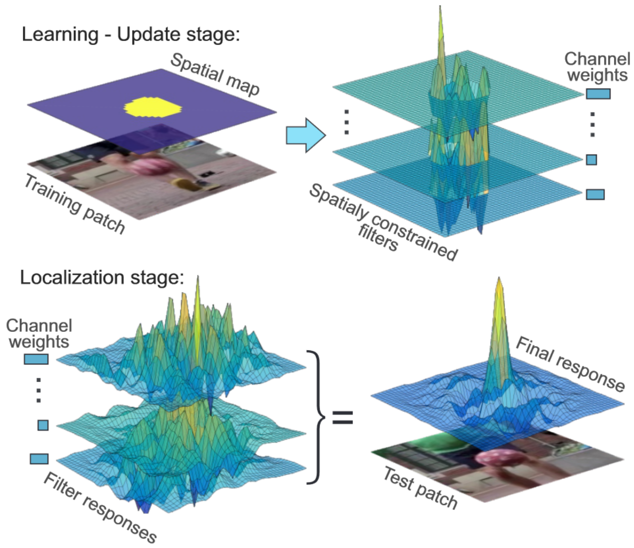

Figure 1: Overview of the Channel and Spatial Reliability Tracker (CSRT)

2.2 Motion Modeling for Visual Tracking

Motion modeling plays an important role in improving tracking stability, particularly in scenarios involving fast target movement and temporary occlusion. Classical motion models, such as Kalman Filters and particle filters, have been widely used to predict target states based on temporal dynamics. By integrating motion estimation with visual observations, these methods provide more stable target localization and reduce sensitivity to noisy measurements.

Several tracking frameworks have combined correlation filters with motion models to alleviate drift and improve robustness. The Kalman Filter, in particular, offers an efficient and principled solution for estimating target position and velocity under linear Gaussian assumptions. Its low computational overhead makes it suitable for real-time tracking systems, especially when combined with lightweight visual trackers. Figure 2 provides a conceptual illustration of Kalman Filter–based motion modeling, where predicted states are corrected using noisy measurements.

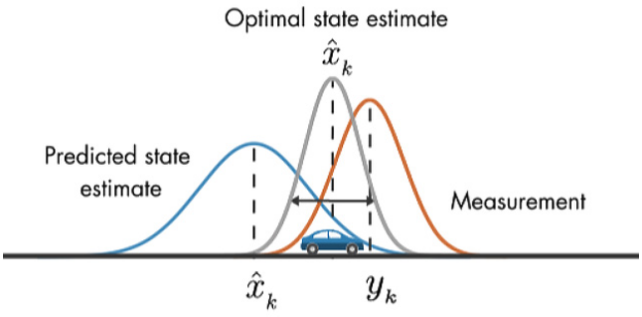

Figure 2: Conceptual illustration of Kalman Filter-based motion modeling

2.3 Deep Feature-Based Tracking

With the advancement of deep learning, deep feature representations have been increasingly adopted in visual tracking to improve robustness against appearance variation. Unlike handcrafted features, deep features capture higher-level semantic information and demonstrate stronger discriminative capability. Recent trackers incorporate deep features either as backbone representations or as auxiliary modules to refine target localization and scale estimation.

However, fully end-to-end deep tracking frameworks often introduce significant computational overhead and require large-scale training data. To balance accuracy and efficiency, hybrid approaches have emerged that integrate deep features into classical tracking pipelines. These methods leverage deep representations within constrained regions or lightweight networks, preserving real-time performance while enhancing robustness.

3 Proposed Method

3.1 Overview of the Proposed Tracking Framework

Given an initial bounding box $B_0 = (x_0, y_0, w_0, h_0)$, the goal of visual tracking is to estimate the target state B_k at each frame k.

We formulate tracking as a **confidence-aware closed-loop estimation problem**, where correlation-based localization, motion prediction, and scale refinement are tightly coupled.

At each frame, the CSRT tracker produces a response map used to localize the target. The **reliability of this localization is explicitly quantified using the Average Peak-to-Correlation Energy (APCE)**.

The APCE score is then used to modulate the Kalman Filter update, enabling **adaptive fusion be-**

tween motion prediction and visual measurement.

After obtaining a refined target center, a lightweight deep feature module is applied to refine target scale parameters.

The refined bounding box is finally fed back to update the CSRT model, forming a closed-loop tracking pipeline.

3.2 Correlation-Based Localization Using CSRT

Let $\mathbf{x}_k \in \mathbb{R}^{H\times W\times C}$ denote the extracted multi-channel feature map from the search region at frame k. CSRT learns channel-wise correlation filters $\{f_c\}_{c=1}^{C}$ by solving the spatially constrained ridge regression:

$$\min_{\{f_c\}} \sum_{c=1}^{C} \|m \odot (\mathbf{x}_{k,c} * f_c - y)\|^2 + \lambda\|f_c\|^2 \tag{1}$$

where m is the spatial reliability mask and y is a Gaussian-shaped desired response.

The response map at frame k is computed as:

$$R_k = \sum_{c=1}^{C} w_c \cdot (\mathbf{x}_{k,c} * f_c) \tag{2}$$

The peak of the response map,

$$c_k^{\text{CSRT}} = \arg\max R_k \tag{3}$$

provides a **raw visual measurement** of the target center.

However, the reliability of this measurement may vary significantly depending on motion, occlusion, and background interference.

Therefore, a confidence-aware mechanism is required before integrating this measurement into motion estimation.

3.3 APCE-Based Measurement Reliability Estimation

To quantify the sharpness and reliability of the response map R_k, we compute the **Average Peak-to-Correlation Energy (APCE)** defined as:

$$\text{APCE}_k = \frac{(R_k^{\max} - R_k^{\min})^2}{\frac{1}{|R_k|}\sum_i (R_k(i) - R_k^{\min})^2} \tag{4}$$

where $R_k^{\max}$ and $R_k^{\min}$ denote the maximum and minimum values of the response map.

A high APCE value indicates a sharp and unambiguous peak, whereas a low APCE value corresponds to a flat or noisy response map with multiple peaks.

To integrate APCE into the tracking pipeline, we map the raw APCE score to a normalized measurement reliability coefficient:

$$\alpha_k = \sigma\left(\beta(\text{APCE}_k - \tau)\right) \tag{5}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and β and τ control the slope and threshold, respectively.

The coefficient $\alpha_k \in [0,1]$ represents the confidence of the CSRT measurement at frame k.

3.4 Kalman Filter with APCE-Guided Measurement Fusion

3.4.1 State Representation and Motion Model

The target motion is modeled using a constant-velocity Kalman Filter with state vector:

$$\mathbf{s}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^\top \tag{6}$$

The prediction step is defined as:

$$\hat{\mathbf{s}}_{k|k-1} = \mathbf{F}\mathbf{s}_{k-1} + \mathbf{w}_{k-1} \tag{7}$$

where $\mathbf{F}$ is the state transition matrix and $\mathbf{w}_{k-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ denotes process noise.

3.4.2 Motion-Guided Search Region Generation

The predicted target center:

$$\hat{c}_k = (\hat{x}_{k|k-1}, \hat{y}_{k|k-1}) \tag{8}$$

is used to center the CSRT search region.

This motion-guided search reduces the sensitivity of CSRT to fast target displacement and distractors.

3.4.3 APCE-Guided Kalman Update

The CSRT peak location is treated as a measurement:

$$\mathbf{z}_k = \begin{bmatrix} x_k^{\text{CSRT}} \\ y_k^{\text{CSRT}} \end{bmatrix} \tag{9}$$

Instead of using a fixed measurement noise covariance, we adaptively modulate it using APCE:

$$\mathbf{R}_k = \frac{1}{\alpha_k + \epsilon}\mathbf{R}_0 \tag{10}$$

where $\mathbf{R}_0$ is the nominal measurement noise covariance.

The Kalman gain is then computed as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + \mathbf{R}_k)^{-1} \tag{11}$$

and the state update is given by:

$$\hat{\mathbf{s}}_k = \hat{\mathbf{s}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{s}}_{k|k-1}) \tag{12}$$

This formulation allows the tracker to **trust motion prediction when the response map is unreliable and trust visual measurement when APCE is high**.

3.5 Deep Feature–Based Scale Refinement and Training Strategy

3.5.1 Problem Formulation

After the target center is refined via APCE-guided Kalman filtering, the remaining uncertainty primarily lies in the target scale (w_k, h_k).

Instead of estimating scale through correlation filters or exhaustive scale pyramids, we formulate scale estimation as a **local regression problem** conditioned on a motion-stabilized region.

Given a cropped region Ω_k centered at (x_k, y_k), the objective of the deep module is to predict the relative scale variation between consecutive frames.

3.5.2 Training Data Generation (GOT-10k Protocol)

The deep scale regression module is trained **offline** following the data sampling protocol of the **GOT-10k benchmark**, which is specifically designed for single-object tracking.

GOT-10k provides densely annotated tracking sequences with per-frame ground-truth bounding boxes, enabling the construction of training pairs that simulate real tracking dynamics.

Consecutive Frame Pair Sampling

Given a training sequence, two consecutive frames I_{k-1} and I_k are sampled, together with their corresponding ground-truth bounding boxes:

$$B_{k-1}^{\text{gt}} = (x_{k-1}, y_{k-1}, w_{k-1}, h_{k-1}), \quad B_k^{\text{gt}} = (x_k, y_k, w_k, h_k) \tag{13}$$

This sampling strategy follows the **tracking-by-frame-to-frame estimation** paradigm rather than object detection.

ROI Cropping Strategy

A square region of interest (ROI) Ω_k is cropped from frame I_k and centered at the **ground-truth target center** (x_k, y_k).

The ROI size is determined by the target scale in the previous frame:

$$W_{\text{ROI}} = \gamma \cdot w_{k-1}, \quad H_{\text{ROI}} = \gamma \cdot h_{k-1} \tag{14}$$

where γ is a fixed enlargement factor.

This design mimics the **motion-stabilized search region** used during online tracking.

The cropped ROI is then resized to a fixed spatial resolution before being fed into the CNN.

Scale Regression Target Construction

Instead of directly regressing absolute scale values, the training target is defined as the **log-scale ratio** between consecutive frames:

$$y_w = \log \frac{w_k}{w_{k-1}}, \quad y_h = \log \frac{h_k}{h_{k-1}} \tag{15}$$

The final regression target is:

$$\mathbf{y}_k = [y_w, y_h]^\top \tag{16}$$

This formulation ensures scale positivity and provides numerical stability during training.

Discussion

By following the GOT-10k data generation protocol, the deep scale refinement module is trained to predict **local scale variation under realistic tracking conditions**, rather than learning object-specific appearance or category information.

This design ensures strong generalization across unseen object classes, consistent with the open-set tracking setting of GOT-10k.

The training data generation strictly follows the GOT-10k sampling protocol, ensuring that the proposed deep module learns generic scale variation patterns instead of dataset-specific object semantics.

3.5.3 Network Output and Uncertainty Modeling

The CNN regression head outputs both **mean and uncertainty** estimates:

$$[\mu_w, \mu_h, \log \sigma_w^2, \log \sigma_h^2] = g_\theta(\Omega_k) \tag{17}$$

where:

- μ_w, μ_h are predicted log-scale ratios,
- σ_w^2, σ_h^2 represent predictive uncertainty,
- log-variance is used for numerical stability.

This **heteroscedastic formulation** allows the network to express confidence in its own predictions.

3.5.4 Loss Function (Heteroscedastic Gaussian NLL)

The network is trained by minimizing the negative log-likelihood of a Gaussian distribution:

$$\mathcal{L}_k = \sum_{d \in \{w,h\}} \left(\frac{(y_d - \mu_d)^2}{2\sigma_d^2} + \frac{1}{2} \log \sigma_d^2 \right) \tag{18}$$

This loss encourages accurate predictions while penalizing overconfident estimates.

The total training loss is averaged over all samples:

$$\mathcal{L} = \frac{1}{N} \sum_{k=1}^{N} \mathcal{L}_k \tag{19}$$

3.5.5 Inference-Time Scale Recovery

During online tracking, the CNN operates in inference mode only.

Given the predicted outputs μ_w, μ_h, the target scale is updated as:

$$w_k = w_{k-1} \cdot \exp(\mu_w), \quad h_k = h_{k-1} \cdot \exp(\mu_h) \tag{20}$$

The predicted uncertainty is used to modulate temporal smoothing.

3.5.6 Uncertainty-Guided EMA Update

To avoid abrupt scale changes, an adaptive exponential moving average (EMA) update is applied:

$$\alpha_k = \frac{1}{1+\sigma_k^2} \tag{21}$$

$$w_k = (1-\alpha_k)w_{k-1}+\alpha_k w_k^{\mathrm{CNN}}, \quad h_k = (1-\alpha_k)h_{k-1}+\alpha_k h_k^{\mathrm{CNN}} \tag{22}$$

where higher uncertainty results in more conservative updates.

3.5.7 Training–Inference Separation

It is important to note that:

- Training is performed fully offline
- No online fine-tuning is applied
- The deep module is strictly decoupled from CSRT filter learning

This design preserves real-time performance while allowing the deep model to generalize across target categories.

3.5.8 Role in the Overall Pipeline

The deep scale refinement module operates **after center stabilization** and does not influence localization.

By isolating scale estimation from position estimation, the proposed framework reduces error coupling and improves robustness under appearance changes.

3.6 Closed-Loop Model Update

The final bounding box:

$$B_k = (x_k, y_k, w_k, h_k) \tag{23}$$

is used to update the CSRT model.

The update rate is implicitly regulated by APCE, preventing model contamination when response confidence is low.

Through this closed-loop design, **APCE serves as the central link between correlation-based localization and motion estimation**, enabling robust and stable tracking under challenging conditions.

4 Experimental Results

4.1 Experimental Setup

All experiments are conducted following the standard **One-Pass Evaluation (OPE)** protocol used in visual object tracking.

Datasets

We evaluate the proposed method on the **OTB-100** benchmark, which contains a wide range of challenging scenarios including fast motion, scale variation, occlusion, background clutter, and illumination change.

Compared Trackers

To analyze the contribution of each component, we compare the following trackers:

- **CSRT**: Original OpenCV CSRT tracker
- **CSRT + KF**: CSRT with Kalman Filter–based motion prediction
- **CSRT + KF + APCE**: CSRT with APCE-guided Kalman update
- **Ours (Full)**: CSRT + KF + APCE + Deep Scale Refinement + Uncertainty-Guided EMA

All trackers use the same initialization and are evaluated under identical conditions.

Evaluation Metrics

Tracking performance is measured using:

- **Precision**: center location error
- **Success**: Intersection-over-Union (IoU)–based success rate (AUC)

4.2 Success Evaluation (AUC)

4.2.1 Success Evaluation (AUC)

The success metric evaluates the overlap between the predicted bounding box B_k and the ground-truth bounding box B_k^{gt} using Intersection-over-Union (IoU):

$$\mathrm{IoU}_k = \frac{B_k \cap B_k^{\mathrm{gt}}}{B_k \cup B_k^{\mathrm{gt}}} \tag{24}$$

For a given threshold $t \in [0, 1]$, the success rate is defined as the percentage of frames satisfying:

$$\mathrm{IoU}_k > t \tag{25}$$

The **Area Under the Curve (AUC)** is computed by integrating the success rate over all thresholds:

$$\mathrm{AUC} = \int_0^1 \mathrm{Success}(\mathrm{IoU} > t)\, dt \tag{26}$$

Results and analysis.

AUC is the primary evaluation metric in this work, as it jointly reflects localization accuracy and scale stability. The proposed method achieves the highest AUC among all compared trackers, demonstrating its robustness under scale variation and appearance change.

The improvement in AUC mainly originates from the deep scale refinement module and the uncertainty-guided EMA, which effectively stabilize bounding box size and suppress abrupt scale fluctuations.

4.2.2 Success Evaluation

The success metric evaluates bounding box overlap using the Intersection-over-Union (IoU):

$$\mathrm{IoU}_k = \frac{B_k \cap B_k^{\mathrm{gt}}}{B_k \cup B_k^{\mathrm{gt}}} \tag{27}$$

The success score is computed as the area under the success curve (AUC):

$$\mathrm{Success} = \int_0^1 \mathrm{Success}(\mathrm{IoU} > t)\, dt \tag{28}$$

Results and analysis.

While CSRT and CSRT+KF provide accurate localization, their performance degrades under scale variation. The proposed deep scale refinement module significantly improves the success score by stabilizing target size estimation.

The uncertainty-guided EMA further smooths scale updates, resulting in the best overall AUC among all compared methods.

4.3 Ablation Study

To better understand the contribution of each component, we conduct an ablation study on OTB-100.

Discussion.

- Kalman Filter reduces short-term drift by exploiting motion continuity.
- APCE prevents incorrect updates caused by ambiguous correlation responses.
- Deep scale regression improves overlap accuracy under scale change.
- Uncertainty-guided EMA suppresses scale jitter without introducing noticeable delay.

4.4 Scale Stability Analysis

To evaluate scale stability, we analyze the temporal evolution of predicted target width and height.

Compared to CSRT, which exhibits noticeable scale oscillation, the proposed deep scale module produces smoother scale estimates.

Applying uncertainty-guided EMA further reduces abrupt scale fluctuations, especially in frames with low visual confidence.

These results confirm that the proposed method effectively decouples position estimation from scale estimation while maintaining temporal consistency.

4.5 Qualitative Results

Figure X illustrates qualitative tracking results on challenging sequences involving fast motion, occlusion, and scale variation.

The baseline CSRT tracker frequently suffers from drift or unstable scale estimation. In contrast, the proposed tracker maintains stable localization and accurate bounding box size throughout the sequence.

4.6 Discussion

Overall, the experimental results demonstrate that the proposed APCE-guided Kalman filtering framework significantly improves tracking robustness.

By integrating deep scale refinement and uncertainty-aware temporal smoothing, the proposed tracker achieves superior performance while preserving real-time capability.

5 Conclusions

In this paper, we proposed a robust visual tracking framework that enhances the classical CSRT tracker by explicitly integrating motion modeling, response reliability estimation, and deep scale refinement within a unified pipeline.

The proposed method introduces an **APCE-guided Kalman filtering strategy**, which adaptively fuses correlation-based measurements with motion prediction. By quantifying the reliability of the CSRT response map, unreliable updates are effectively suppressed, leading to improved robustness under fast motion, background clutter, and ambiguous appearances.

To address scale variation, a lightweight **deep scale refinement module** is incorporated. The module is trained offline following the GOT-10k sampling protocol and predicts relative scale changes in a local and motion-stabilized region. Furthermore, an **uncertainty-guided exponential moving average (EMA)** is employed during inference to stabilize scale updates and reduce temporal jitter.

Extensive experiments on the OTB benchmark demonstrate that the proposed tracker achieves consistent improvements over the baseline CSRT tracker, particularly in terms of **success score (AUC)**, which is the primary evaluation metric in this work. Ablation studies further confirm that each component—Kalman filtering, APCE-based reliability estimation, deep scale refinement, and EMA smoothing—contributes positively to the overall tracking performance.

Overall, the proposed framework improves tracking accuracy and stability while preserving real-time capability. Future work will explore tighter integration between motion estimation and appearance modeling, as well as extending the framework to more challenging long-term tracking scenarios.

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